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What Rich Interfaces Add to Battery-Free Design
A Matched Digital Study with Strong Static Controls

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Abstract

Battery-free design requires decisions about energy arrival, startup, event capture, reset and delivery. We compare information and observation interfaces on four fixed digital tasks with three formal repetitions per task and condition. The average-energy arm achieves 8 PASS and 4 FAIL; complete-input, rich-static and executable-response arms each achieve 12 PASS. Rich-static therefore remains a successful strong control, and executable access shows no service advantage on these tasks. Prospective live observation (E5) achieves 7 PASS and 5 INCOMPLETE, with every incomplete group retained in the denominator. Its successes comprise six selections of supplied fields and one constructed relation; none adds a physical measurement. An equal-information JSON/table comparison achieves 12 PASS and 65 provider requests in each arm. Four fresh Agent groups select delivery across a reset, but all reuse one historical digital response, rather than constituting four new physical experiments. These finite public development results establish operational interface behavior and its failures. They do not establish interface superiority over complete static information, new physical knowledge, hardware qualification, transfer or recursive self-improvement.

1 Question and evidence scope

An energy total alone does not specify when a device becomes ready, whether an event is captured, whether pending state survives a reset, or whether a reply reaches a receiver before its deadline. A richer interface might expose those facts as supplied records, an executable response, or a model-proposed typed observation. The relevant comparison must include complete static information: otherwise a successful interactive arm could merely recover facts withheld from its baseline.

This study asks whether richer or executable battery-free information improves complete service under a matched finite budget. The strong version of the registered hypothesis requires value beyond complete static inputs. The present results do not support that stronger claim. The information ablation misses service in four reset-task runs, while all strong static and executable arms pass. Live observation demonstrates seven prospective source-to-consumer chains but also five admission failures. We report these as distinct outcomes, without treating a missing final execution as a failed physical design.

The evidence is the completed Phase 3 public development matrix and its frozen protocol, repeat decision and source-bound ordered-event receipts [1, 2, 3, 4, 5]. The four task identifiers denote four fixed inputs. Repetitions, method owners, cached responses and selected witnesses do not create additional tasks. All scientific conclusions concern the fixed ideal digital apparatus and one preregistered synthetic DC derivative. Physical qualification remains UNKNOWN.

2 Frozen apparatus and matched protocol

2.1 Four fixed tasks and a complete-service outcome

The first three tasks use the same versioned digital transient apparatus: `b-pulse-fast`, `b-volatile-reset` and `b-latched-reset`. Their design variables are capacitance and the earliest allowed transmission time, within task-specific legal ranges. Pulse-fast is a static-sufficient control. The reset cases expose the timing of capture, energy loss and later delivery under volatile or ideal passive-latch state. A complete-service PASS uses the registered independent receiver and event-order check. FAIL means an admitted final execution failed that check. INCOMPLETE means no admitted final execution; it is retained as a third outcome, never removed or replaced.

The fourth task, `clite-query-window-derived-v1`, is a preregistered single-window derivative of proposed C-lite requirements [7]. It prescribes 20 μW DC input over 30 s, 1 μW idle power, zero initial energy and a 50 μJ reply. A query arrives at 2 s; the reply deadline is 7 s and maximum data age is 5 s. Capacitance is limited to 25–100 μF, corresponding to at most 200 μJ at the ideal 2 V ceiling. Zero ESR, unit harvest efficiency and zero boot energy/delay are explicit digital assumptions. They are not measured component properties. The registered gates require query issuance, capture, the matching token, causal ordering, deadline and freshness. Fixed sequence/key/event value 1 represents one virtual query token. It establishes neither a multi-query nonce protocol nor security, RF behavior or qualified hardware.

2.2 Conditions and information boundaries

All arms receive the same task, legal design space, declared assumptions and units. Average-energy starts with the coarse source integral and average power. Rich-static and complete-raw both include every supplied input record and the full model definition. Their equality in available facts is an admissible negative control; "complete-raw" is the registered arm name, not a claim that the model received the full coupled raw physical trace. That trace remains host-side for audit. We call this arm complete-input in the tables to keep the boundary visible.

Executable-response receives the same complete inputs plus candidate-specific public energy ledgers, activity and packets. Live E5 instead starts coarse and actually acquires sources through a generic interface. It can select a supplied field or construct a bounded expression, declare its domain and units, submit a typed consumer policy, and obtain independent qualification before final service checking. No target expression or answer-bearing field relevance is prefilled. Qualification checks the declared source relation; it does not by itself prove service, novelty or the causal necessity of that observation.

E4 compares the rich-static packet in JSON with a typed JSON-Pointer table. All 12 formal pairs have identical decoded initial packets and identical tools. The table preserves containers, types, empty values and nested structure, without adding summaries or answers. Later source-acquisition responses use common JSON in both arms. Thus E4 isolates initial serialization only, rather than the representation of an entire interactive trajectory.

2.3 Accounting unit, freeze and cache

The complete eight-condition matrix has 112 unique groups and 140 actual episodes: 96 formal groups (four tasks $\times$ three repetitions $\times$ eight conditions; 120 episodes) and 16 pilot groups (one task $\times$ two repetitions $\times$ eight conditions; 20 episodes). A group includes every charged parent method-proposal episode when one is present. This paper's six displayed interface arms contain 72 formal and 12 pilot groups. Complete-raw is also the method study's fixed-H cell; rich-static is also E4 JSON. Each shared cell is counted once, not redispatched or added again to the global totals.

The two pilots on volatile-reset precede the formal comparison. Under the registered one-time rule, instability of designs/outcomes selects three formal repetitions, rather than two. The repeat freeze records exactly one selection event and forbids adaptation afterward [4]. A failed group is not automatically retried or replaced; the fixed registered repetitions are separate groups. Receipts report the model as `deepseek-flash` throughout

this matrix. These repetitions describe behavior on four used inputs, not independent samples of new task families or a statistical acceptance test.

Every group has a total allowance of 24 provider requests, 40 tool calls, 1,200 s and at most two distinct design exposures, including checks, response queries and confirmation. E5 acquisition, qualification and consumption receive no extra design or total budget. Where used, a parent H proposal has an eight-request, 400 s sublimit within the group accounting. The campaign caps are 3,000 requests, USD 200 in settled conservative provider estimates and 134,400 summed episode seconds. Paid/new dispatch is closed for paper preparation.

Every arm starts from the same hash-bound immutable cache: twelve Phase 2 exact-input responses and one registered derivative baseline. Private overlays isolate group outputs. A physical cache hit consumes logical exposure and does not reveal another arm's output before a query or enlarge the design cap. Here "physical response" is the cache accounting term for an underlying digital execution, not evidence of laboratory hardware. Public development feedback and the independent final check retain their registered boundaries.

3 Formal results and separate pilot evidence

3.1 Strong static controls saturate complete service

Table 1 retains all twelve formal groups per arm. Average-energy has 8 PASS and 4 FAIL. Rich-static, complete-input, executable-response and E4 table each pass all twelve. E5 has 7 PASS and 5 INCOMPLETE; there is no admitted service FAIL in that arm. Those five incomplete groups cannot be interpreted as five simulated physical failures or omitted to report a conditional 7/7 success rate.

Table 1: Formal interface results. Each row is four fixed tasks with three repetitions each; P/F/I denote PASS/FAIL/INCOMPLETE. Requests include failures.

Condition	P	F	I	Requests	Mean/group
Average-energy	8	4	0	75	6.2500
Rich-static / E4 JSON	12	0	0	65	5.4167
Complete-input (complete-raw)	12	0	0	64	5.3333
Executable-response	12	0	0	66	5.5000
Live observation (E5)	7	0	5	80	6.6667
E4 table	12	0	0	65	5.4167

The task breakdown in Table 2 locates the coarse arm's observed deficit in the two reset tasks: each has one PASS and two FAIL, while pulse-fast and the C-lite derivative each have three PASS. The four strong arms each pass all three repetitions on every task. This finite contrast is consistent with the value of detailed timing/state information beyond the coarse summary. It provides no service advantage for executable-response over the complete static baseline, and no evidence that live E5 surpasses that baseline. Saturation on four public tasks also prevents a conclusion that the strong interfaces would be equivalent on harder or unseen tasks.

Table 2: Per-task formal counts. Each entry is P/F/I out of three; the strong column applies separately to rich-static, complete-input, executable-response and E4 table. Repetitions do not enlarge the four-task denominator.

Fixed task	Average-energy	Live E5	Each strong arm
Pulse-fast	3/0/0	2/0/1	3/0/0
Volatile-reset	1/2/0	2/0/1	3/0/0
Latched-reset	1/2/0	1/0/2	3/0/0
C-lite query derivative	3/0/0	2/0/1	3/0/0

3.2 Pilots are development evidence

Table 3 reports the two pilot repetitions on volatile-reset separately. Average-energy has one PASS and one FAIL; every other displayed arm has two PASS. The two E5 pilot successes include one construction and one selection. These observations helped determine the frozen repeat count; they are not pooled into the formal twelve-run denominator or treated as extra tasks.

Table 3: Separate pilot results: one fixed task and two repetitions per arm. The full eight-arm pilot contains sixteen groups and twenty actual episodes.

Condition	P	F	I	Requests
Average-energy	1	1	0	15
Rich-static / E4 JSON	2	0	0	11
Complete-input (complete-raw)	2	0	0	10
Executable-response	2	0	0	10
Live observation (E5)	2	0	0	11
E4 table	2	0	0	12

4 What the prospective E5 chains establish

All seven formal successes have an actual prospective submitted observation, external source qualification and recorded typed consumer use. Six select already supplied fields; one constructs a relation. Five use a component record and two an excitation record. Six change the initial design and one leaves it unchanged. Every successful final design passes the independent service check, but typed consumption and design change do not establish that the observation was necessary or unavailable to the complete static arm. All seven are based on supplied data; new physical measurements number zero.

Table 4: Disposition of all twelve formal E5 groups. Failure categories count groups, not repeated rejected tool actions. Every row remains in the denominator.

Disposition	Groups	Final service status
Qualified selection, consumed	6	PASS
Qualified construction, consumed	1	PASS
Source-unit binding rejected	4	INCOMPLETE; no admitted final execution
No typed consumer match	1	INCOMPLETE; no admitted final execution

Four groups record `SOURCE_UNIT_UNAVAILABLE` in the original gateway events [8]. This code concerns the submitted source/field/domain unit binding; it does not establish that a physical quantity does not exist. For example, the pulse-fast proposal selects `/packets/0/application_bytes`, which *is* unit-mapped in its acquired saved-activity record, while domain constraints reference deadline, TX-duration and readiness fields absent from that same source's unit map. A latched-reset proposal instead selects a packet field from the model-law source, whose mapped fields are model identity and Python definition. These original submissions show source-local binding mismatches. The generic error does not identify the first rejected term, and no repaired proposal is evaluated here.

The remaining incomplete C-lite group records `NO_TYPED_CONSUMER_MATCH`. Its submitted equality expression has Boolean output, while its consumer branches match the numeric durations 0.002 and 0.001. This syntax-level diagnosis uses the original proposal and log; it is not a qualifier or consumer rerun. Other syntax and valid-proposal-cap errors in these groups remain charged history, rather than additional independent failures or free rescue attempts.

The earlier Phase 2 actual episode remains a failed live E5 attempt. Its separately labeled post-hoc adapter chain reused an already generated corrected call; it is neither a prospective Phase 3 success nor a replacement

for that failure [9]. The seven formal chains reported here are actual prospective submissions under the later registered protocol. This separation preserves the distinction between an engineering repair demonstration and prospective operational reliability.

5 Equal-information serialization and ordered witnesses

5.1 E4 gives no observed service or request advantage

Both E4 arms achieve 12/12 formal PASS and exactly 65 provider requests, or 5.4167 per group. Equality of decoded initial packets and tools holds for all twelve formal pairs. Table 5 gives provider cost and wall time only as descriptive receipts. Equal service and request totals do not prove general equivalence; the cost and wall differences do not establish a format advantage. The study neither isolates all determinants of latency/cost nor tests a representation change in later common-JSON acquisitions.

Table 5: E4 formal descriptive accounting. Dollars are provider-reported settled conservative estimates, not an invoice audit. Wall time is summed group time; no causal format-speed or format-cost claim is made.

Initial representation	PASS	Requests	USD	Wall (s)
JSON (rich-static)	12/12	65	0.4267200	676.5
Typed table	12/12	65	0.5145681	611.4

5.2 Four fresh selections reuse one historical response

Complete-service delivery and state retention across a reset are distinct observations. A retention witness requires capture of the same event before a brownout and its first valid delivery afterward. A reset after delivery, or a PASS Boolean alone, is insufficient. The source-bound initial controls are classified as delivery before any reset (pulse-fast), no valid event delivery (volatile-reset), and pending-event retention across reset (latched-reset) [5]. These labels analyze existing responses without a new scientific execution.

Four formal Agent groups select the across-reset outcome: two complete-input (fixed-H) groups, one proposed-method group, and one E4 table group. They are four fresh Agent choices but share one distinct historical digital execution key. All choose $C = 10$ μF and an earliest TX time of 45 ms. Table 6 records their common ordered witness. The fixed ideal passive-mechanical latch supplies the retention mechanism; these choices neither invent a memory mechanism nor qualify a physical implementation. All four have physical qualification UNKNOWN. The witness audit creates zero new scientific executions.

Table 6: One historical digital response reused by four formal selections. Capture ends before brownout, and first valid reception follows brownout.

Event	Time (ms)	Source-bound interpretation
Capture interval	3.7–7.9	Same task event 1 captured before reset
Brownout	12.1	Pending captured event retained by the ideal latch
Earliest allowed TX	45.0	Selected design parameter
First valid receive	46.9	Same event first delivered after reset

Most other formal successes deliver before any reset. All twelve rich-static, all twelve executable-response and all seven successful E5 groups are in that category. Such service successes do not demonstrate survival of pending state across a reset. Four selected groups and one underlying execution key must not be reported as four new physical experiments or four independent retention mechanisms.

6 Accounting, interpretation and next gate

The full eight-condition campaign records 771 provider requests, all SETTLED, with no new UNKNOWN usage. Its settled conservative estimate is USD 5.4478605 and summed episode wall time is 8914.3 s. This

accounting includes the two method-proposal arms and their parent episodes, not only the six interface rows. It is provider-reported accounting without an invoice audit; Codex assistance and full end-to-end cost remain UNKNOWN. The current paper rendering adds no provider request, scientific execution or native dispatch.

The observed coarse-information deficit is task dependent. Strong static inputs already provide enough information to reach complete service on all four fixed tasks. An executable interface may still provide useful audit structure, but this study does not demonstrate an additional service benefit. E5's seven successful chains establish that acquired source data can be admitted, qualified and consumed prospectively. The five incomplete chains expose the operational cost of that route under the same total allowance. Its six field selections and one construction do not establish autonomous discovery of an unseen physical relation.

The fixed tasks, small repeat count, one reported model, development exposure and saturated controls limit generalization. Synthetic DC input does not supply RF field measurements, conversion/load-response waveforms, calibrated startup or leakage, clock uncertainty or actual PHY reliability. A full host-side trace and an independently checked digital PASS therefore remain separate from physical qualification.

The preregistered Phase 4 selection rule promotes a proposed procedure only with more task-level complete-service passes than both fixed-H and reflection, no lost task and no higher mean group request count [6]. All three formal task-success vectors are $(3, 3, 3, 3)$; proposed-method requests average 10.5833 per group, versus 5.3333 for fixed-H and 9.7500 for reflection. The recorded decision is to retain unchanged base H. The two reserved within-model parameter-generalization tasks remain unexecuted here and require separate admission. Historical controls do not become unseen tasks. Nothing in Phase 3 establishes global transfer of H or recursive improvement.

6.1 Reproducible manuscript binding

`render_phase3.py` reads the seven authoritative Phase 3 JSONs and verifies their group, outcome, accounting and hash relationships before generating this methods/results fragment. It also reads the five original failed-group gateway logs for the bounded diagnostics above, without opening compressed provider archives. It invokes no simulator, model, qualifier, consumer or service checker. `phase3-binding.json` binds exact source bytes, both TeX sources, the renderer, shared layout/logo and final PDF. The paper-final receipts record compilation, per-page visual inspection and cleanup of this build's temporary files. This validates the manuscript's presentation and source binding; it is not a new scientific replication or a full audit of the historical campaign.

References

[1] S0. Phase 3 authoritative main tables and data dictionary. `main-tables.json`, 20 October 2026.

[2] S0. Source-bound actual group comparison. `actual-comparison.json`, schema `bf-phase3-actual-comparison-v1`.

[3] S0. Controlled matrix protocol. `protocol.json`, identifier `phase3-bf-matched-v1`.

[4] S0. One-time formal repeat decision. `formal-repeat-freeze.json`.

[5] S0. Ordered initial controls and actual selected witnesses. `ordered-initial-controls.json` and `ordered-selected-witnesses.json`.

[6] S0. Frozen follow-up selection rule and task reservation. `stage4-selection-freeze.json`.

[7] B. Frozen interface conditions and synthetic case derivation. `configs/studies/phase3-20261020/interfaces/conditions.json` and `derived-case.json`. Actual execution status comes from the later matrix, not the preparation files' historical "not executed" status.

[8] S0. Original Phase 3 E5 gateway event records for the five incomplete groups; exact paths and SHA-256 bindings in `phase3-binding.json`.

[9] B/S0. Frozen Phase 2 actual-attempt and post-hoc boundaries. `phase2-binding.json`; neither record is counted as a Phase 3 success.